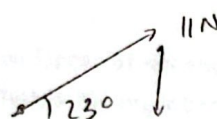


Name: _____

Date: May 22, 2025

For all questions, round lengths to the nearest tenth and angles to the nearest degree.

1. A force vector has a magnitude of 11 N and makes an angle of 23° with the x-axis. What is the magnitude of the vertical component?


 $\sin \theta = \frac{O}{H}$
 $O = H \sin \theta$
 $O = 11 \text{ N} (\sin 23)$
 $O = 4.2980 \text{ N}$
 the vertical comp. is 4.3 N (2)

2. An angle of 25° is formed when vectors $\vec{a}$ and $\vec{b}$ are placed tail-to-tail. If $|\vec{b}| = \frac{7}{3}$ and $\vec{a} \cdot \vec{b} = 12.7$, determine $|\vec{a}|$?

$a \cdot b = |a||b| \cos \theta$
 $|a| = \frac{a \cdot b}{|b| \cos \theta}$
 $|a| = \frac{12.7}{(\frac{7}{3}) \cos 25}$
 $|a| = 6.055$
 $|a| = 6.0$ (2)

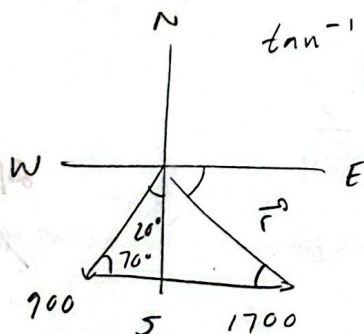
3. Can three forces of magnitudes 4 N, 5 N and 10 N be in equilibrium? Justify your response.

$\sqrt{4^2 + 5^2} < 10 \Rightarrow$ cannot be a triangle
 

$4 + 5 < 10 \Rightarrow$ cannot be a straight line
 

No because 4 and 5 are less than 10 so cannot be collinear opposites and $\sqrt{4^2 + 5^2}$ is less than 10 so cannot be

4. Two forces are applied to an object, $\vec{F}_1 = 900 \text{ N } [S20^\circ W]$ and $\vec{F}_2 = 1700 \text{ N } [E]$. Find the resultant and equilibrant vector's magnitude and direction. Include a vector diagram.



$\tan^{-1} \left(\frac{895.72}{1392.18} \right) = 31^\circ$

in a triangular vector formation, so there is no way for them to be in equilibrium. (4)

$R = \sqrt{900^2 + 1700^2 - 2(900)(1700) \cos 70}$
 $= 1628.93$

$\frac{\sin \theta}{900} = \frac{\sin 70}{1628.93}$
 $\theta = 31^\circ [E 31^\circ S]$

$x \text{ comp: } 1700 - 900 \sin 20$
 $= 1392.18 [E]$

$y \text{ comp: } 900 \cos 20$
 $= 895.72 [S]$

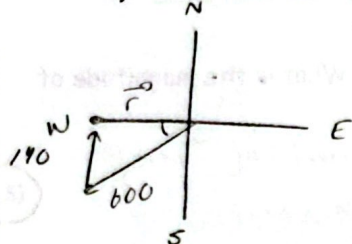
$F = \sqrt{1392.18^2 + 895.72^2}$
 $= 1628.93$

Resultant: 1628.9 N [E 31° S]

Equilibrant: 1628.9 N [W 31° N]

5. An airplane has an airspeed of 600 km/h. The pilot wishes to fly to a destination that is 900 km due west from the plane's present location. There is a 140 km/h wind from the south.

a) Determine the resultant ground velocity. Include a vector diagram.



$$\begin{aligned} r &= \sqrt{600^2 - 140^2} \\ &= 20\sqrt{851} \\ &= 583.4 \text{ km/h [W]} \end{aligned}$$

(2)
1.5

b) In which direction does the plane need to fly to reach the destination?

$$\begin{aligned} \theta &= \sin^{-1}\left(\frac{O}{H}\right) \\ &= \sin^{-1}\left(\frac{140}{600}\right) \\ &= 13.49^\circ \end{aligned} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \text{ [W } 13.5^\circ \text{]}$$

(1)

c) How long will it take to reach the destination?

$$\begin{aligned} t &= \frac{d}{v} \\ &= \frac{900 \text{ km}}{583.4 \text{ km/h}} \\ &= 1.54 \text{ h} \\ &= (1.54 \text{ h}) \left(\frac{60 \text{ min}}{1 \text{ h}} \right) \\ &= 92.5 \text{ min} \end{aligned}$$

(1)

6. The angle between the vectors $\vec{a} = (3, 4, 0)$ and $\vec{b} = (x, 0, 5)$ is $\cos^{-1}\left(\frac{3}{\sqrt{34}}\right)$. Determine the value of x .

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$

$$(3, 4, 0) \cdot (x, 0, 5) = \sqrt{3^2 + 4^2 + 0^2} \cdot \sqrt{x^2 + 0^2 + 5^2} \cdot \frac{3}{\sqrt{34}}$$

$$3x + 4(0) + (0)(5) = 5 \cdot \frac{3}{\sqrt{34}} \cdot \sqrt{x^2 + 5^2}$$

$$(3x)^2 = \left(\frac{15\sqrt{34}}{\sqrt{34}} \cdot \sqrt{x^2 + 5^2} \right)^2$$

$$9x^2 = \frac{225}{34} (x^2 + 5^2)$$

$$9x^2 = \frac{225x^2}{34} + \frac{5625}{34}$$

$$\frac{81x^2}{34} = \frac{5625}{34}$$

$$x = \pm \frac{625}{9}$$

$$\boxed{x = \pm \frac{625}{9}}$$

$$\boxed{x = \pm 69.4}$$

2.5
(3)

6